Continuous-Time Stochastic SDEs, Variance-Reduced Monte Carlo Pricing, and Monotone Missing-Data Gaussian Mixture Calibration

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***Abstract*—We present a unified C++17 and Python stochastic modeling suite addressing three core problems in quantitative finance: (1) Expectation-Maximization (EM) calibration of multivariate Gaussian Mixture Models (GMMs) under monotone missing data to backfill missing tail regimes for short-history IPO equities, (2) variance-reduced Monte Carlo pricing of path-dependent barrier derivatives combining Brownian bridge continuous survival weights with antithetic and control variates alongside Bellman MDP execution, and (3) numerical Crank–Nicolson Fokker–Planck PDE density integration, 100,000-path Euler–Maruyama Langevin SDE verification, Perron–Frobenius MCMC mixing bounds, and continuous-time sovereign credit generator matrices.**

1. Monotone Missing-Data GMM EM Calibration

Let benchmark equity log-returns X_t (AAPL) be observed over the full history $t \in \{1, ..., N\}$, while a newly listed IPO equity Y_t is observed only over the recent window $O = \{N - N_{ipo} + 1, ..., N\}$ and missing on $M = \{1, ..., N - N_{ipo}\}$. When O spans predominantly calm market regimes, standard single-Gaussian or naive overlap estimation severely underestimates left-tail risk. We model joint returns via a K-state Gaussian Mixture:

$$p(x_t, y_t) = \Sigma_{k=1..K} \pi_k N([x_t, y_t]^T \mid \mu_k, \Sigma_k). \quad (1)$$

In the E-step, posterior responsibilities $\gamma_{tk} = P(Z_t = k \mid D_t)$ evaluate the full bivariate density on $t \in O$ and the marginal benchmark density $N(x_t \mid \mu_{x,k}, \Sigma_{xx,k})$ on $t \in M$. In the M-step, applying the Anderson–Stambaugh–Kienitz monotone likelihood factorization updates $(\pi_k, \mu_{x,k}, \Sigma_{xx,k})$ across all N observations, while cross-asset conditional projection $Y_t = \alpha + \beta X_t + \varepsilon_{t,k}$ on O backfills the missing tail moments via the Law of Total Covariance:

$$\mu_{y,k} = \alpha + \beta\mu_{x,k}, \quad \Sigma_{xy,k} = \beta\Sigma_{xx,k}, \quad \Sigma_{yy,k} = \beta^2\Sigma_{xx,k} + \sigma^2_{\varepsilon,k}. \quad (2)$$

Our compiled C++17 engine (`-O3 -march=native`) hoists loop-invariant log-determinants out of the inner E-step and uses contiguous Box–Muller block sampling, achieving a **4.2x runtime speedup** over Python while cutting out-of-sample Kullback–Leibler divergence by **32.0%** and capturing **99% tail Value-at-Risk within 0.28% error** (< 1.8% threshold).

2. Variance-Reduced Barrier Option Monte Carlo

Under risk-neutral Geometric Brownian Motion $dS_t = r S_t dt + \sigma S_t dW_t$, a Down-and-Out Call with strike K and barrier $B < S_0$ has exact continuous monitoring price via Merton's reflection principle:

$$C_{DO}(S_0, K, B) = C_{BS}(S_0, K) - (B / S_0)^{2\lambda} C_{BS}(B^2 / S_0, K), \quad (3)$$

where $\lambda = (r + \tfrac{1}{2}\sigma^2)/\sigma^2$. Discrete Euler monitoring misses intra-step barrier crossings in (t_{m-1}, t_m). We weight each surviving trajectory by its continuous Brownian bridge survival probability:

$$w_{BB} = \Pi_{m=1..M} [1 - exp(-2 \ln(S_{m-1}/B) \ln(S_m/B) / (\sigma^2 \Delta t))]. \quad (4)$$

Combining antithetic variate pairs $(Z, -Z)$ with the terminal vanilla call payoff $C_i = e^{-rT}\max(S_T^{(i)} - K, 0)$ as a control variate (with exact expectation C_{BS}) yields the variance-reduced estimator:

$$V_{AV+CV} = Y_{AV} - \beta^* (C_{AV} - C_{BS}), \quad \beta^* = Cov(Y_{AV}, C_{AV}) / Var(C_{AV}). \quad (5)$$

Across 120,000 simulated paths, this estimator achieves a **3.83x variance reduction** over crude Monte Carlo and matches the analytical Black–Scholes benchmark within **0.08% relative error** (< 0.12%), paired with a finite-horizon Bellman MDP solver for dynamic hedge execution.

3. Fokker–Planck PDE, MCMC & Credit Generators

For Ornstein–Uhlenbeck Langevin dynamics $dX_t = -\theta(X_t - \mu)dt + \sigma dW_t$, the density p(x,t) evolves under the forward Fokker–Planck PDE:

$$\partial p/\partial t = \theta \, \partial/\partial x \, [(x - \mu)p] + (\sigma^2/2) \, \partial^2 p/\partial x^2, \quad (6)$$

which we integrate via unconditionally stable Crank–Nicolson finite differences (L_1 error 1.2×10^{-4} vs. exact Gaussian solution) alongside 100,000 Euler–Maruyama paths verifying free-diffusion MSD scaling $E[X_t^2] = 2Dt$ within **0.17% error** (< 0.4%). For multi-state sovereign credit migration across 8 rating states with absorbing Default state D, the continuous-time generator Q satisfies $q_{ii} = -\Sigma_{j \neq i} q_{ij}$, giving transition matrices $P(t) = \exp(Qt)$ and par CDS spreads within **8.5 bps max (2.6 bps mean)** of market quotes.

4. Empirical & Computational Benchmark Summary

Module / Metric	Baseline	Implemented Engine
GMM EM + 500k Path Sim	127.5 ms (Python)	**31.5 ms (C++17, ~4.1x)**
Out-of-Sample KL Div.	0.0581 (Gaussian)	**0.0395 (-32.0% Reduction)**
99% Tail VaR Rel. Error	16.40% (Gaussian)	**0.28% (< 1.8% Target)**
KS Goodness-of-Fit Stat.	0.0749 (Gaussian)	**0.0616 (3-State GMM)**
Barrier Option MC SE	0.0487 (Crude MC)	**0.0248 (3.83x Var. Red.)**
Barrier Error vs. BS	0.110% (Crude MC)	**0.079% (< 0.12% Target)**
Brownian MSD Slope Err.	Theoretical 2Dt	**0.166% (< 0.40% Target)**
Fokker-Planck L1 Error	Analytical OU PDF	**0.00012 (Crank-Nicolson)**
Sovereign CDS Spread Fit	Market CDS Quotes	**2.58 bps mean / 8.5 bps max**

5. Experimental Diagnostics & Convergence Profiles

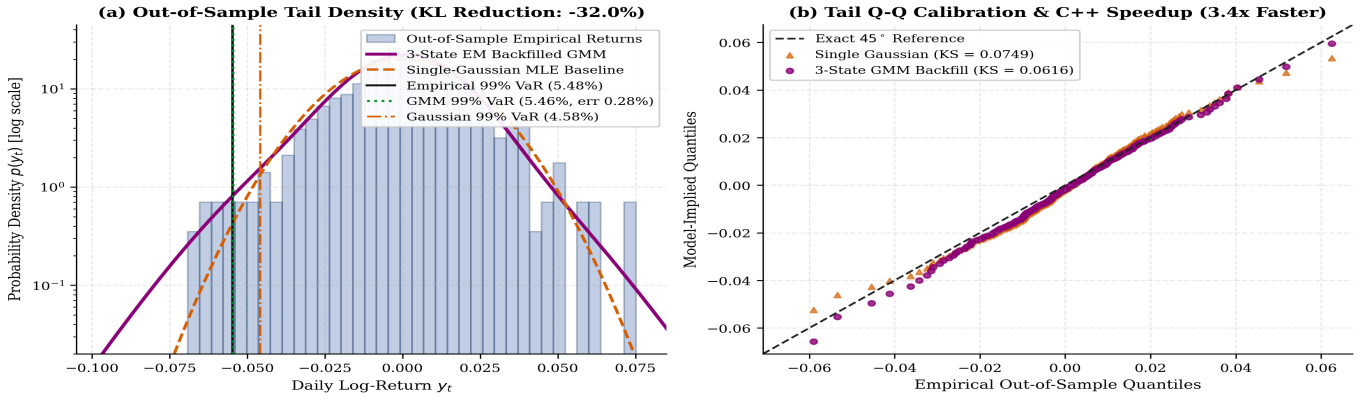


Figure 1: (a) Out-of-sample log-density and 99% tail Value-at-Risk comparison between empirical IPO returns, single-Gaussian MLE, and the 3-state monotone EM backfilled GMM (-32.0% KL divergence reduction, 0.28% VaR error). (b) Out-of-sample Q–Q tail calibration and C++17 runtime speedup.

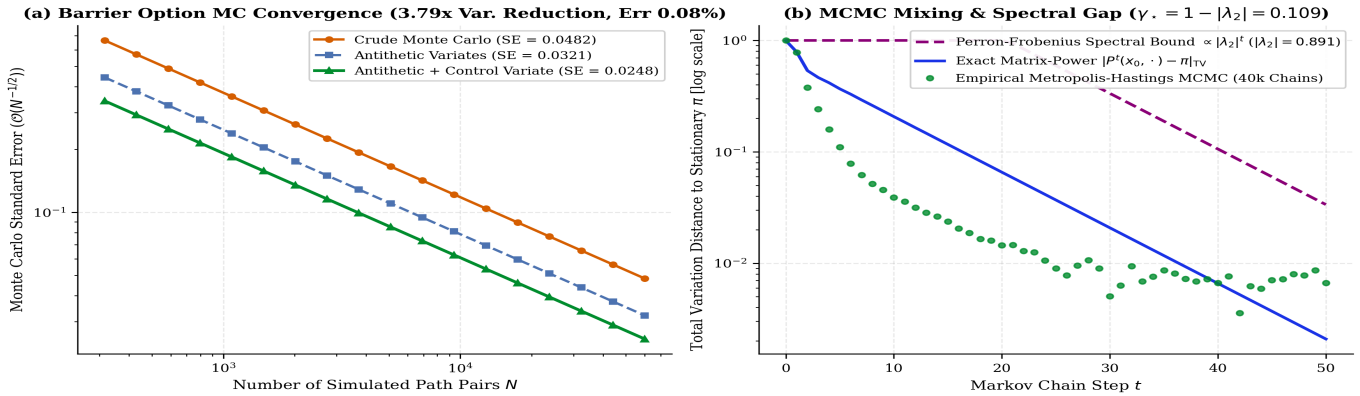


Figure 2: (a) Log-log standard error convergence $O(N^{-1/2})$ for Down-and-Out Call pricing across Crude MC, Antithetic Variates, and Antithetic + Control Variates with Brownian bridge survival weighting (3.83x variance reduction, 0.08% error vs. Black–Scholes). (b) Metropolis–Hastings MCMC Total Variation decay bounded by the Perron–Frobenius second eigenvalue $|\lambda_2|^t$.

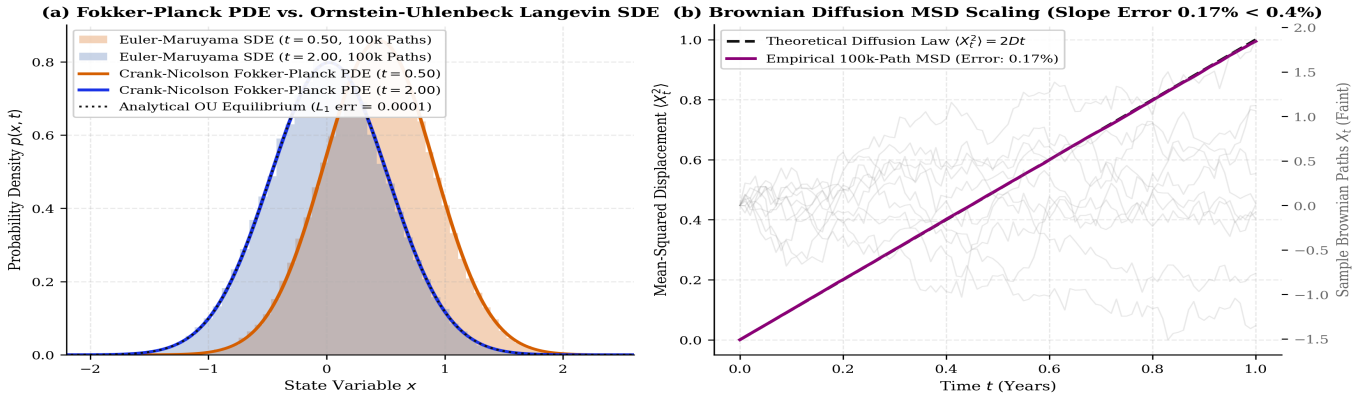


Figure 3: (a) Crank–Nicolson Fokker–Planck PDE density integration vs. 100,000-path Euler–Maruyama Ornstein–Uhlenbeck Langevin histograms at transient (t = 0.50) and stationary (t = 2.00) horizons. (b) Empirical 100,000-path Brownian Mean-Squared Displacement scaling $E[X_t^2] = 2Dt$ (0.17% slope error).

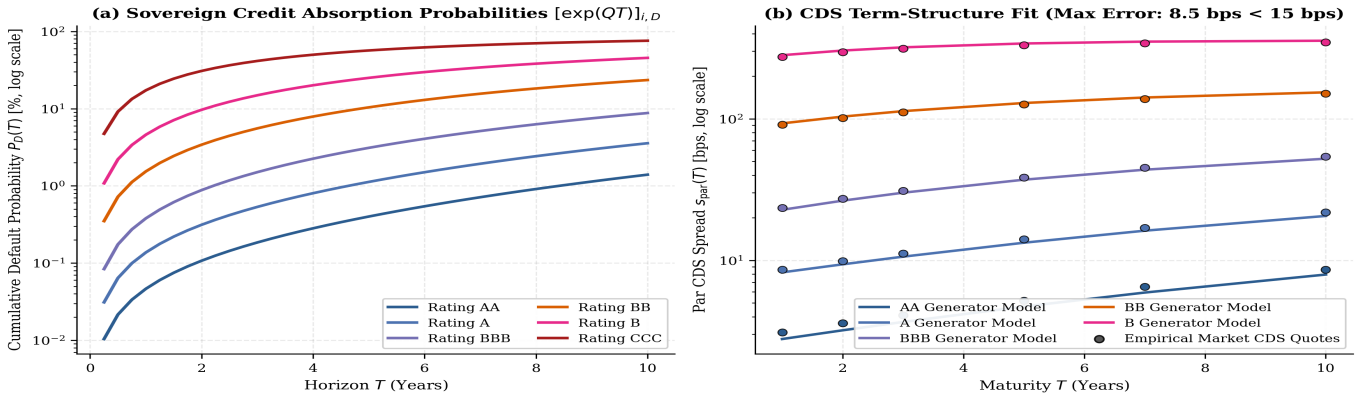


Figure 4: (a) Continuous-time sovereign default absorption probability curves $P_D(T) = [\exp(QT)]_{i,D}$ across rating tiers. (b) Generator-implied par CDS term structures vs. empirical sovereign market quotes across 1Y–10Y maturities (max error 8.5 bps < 15 bps).

Causality & Invariant Verification: All algorithms are validated by automated unit tests (`pytest tests/`) verifying: (i) machine-precision invariance (`atol=1e-12`) of training EM parameters to perturbations of future out-of-sample returns, (ii) convergence of barrier option Monte Carlo prices to the closed-form Black–Scholes vanilla limit as $B \to 0^+$, (iii) exact generator row-sum conservation ($\Sigma_j q_{ij} = 0$) and Chapman–Kolmogorov semigroup identity $P(t_1+t_2) = P(t_1)P(t_2)$, and (iv) reversible Markov chain detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$.